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Mechanism and Performance of SiO_2 Nanoparticle Enhanced Carbonated Polymeric Nanofluids for EOR and CO_2 Sequestration

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Abstract

This study aimed to examine the effects of SiO_2 nanoparticles (NPs) on enhanced oil recovery using carbonated polymeric nanofluids. The primary objectives of this study were to reduce interfacial tension (IFT), modify the wettability, and improve the sweep efficiency by adjusting the viscosity of the slug. The most recent development in this field utilizes carbon dioxide dissolved in a slug for EOR, which offers the dual benefits of geological CO_2 trapping and decarbonization. This investigation examined the use of SiO_2 nanoparticles, which were subsequently evaluated for their microstructural characteristics and particle size distribution. The suitability of the substances for reservoir conditions was determined by assessing their interface and rheological properties under both ambient and high-pressure conditions. Modification of the interfacial and rheological characteristics of the prepared solutions can aid in optimizing the chemical concentration. The integration of nanoparticles into a typical oil field PAM polymer serves as a mobility-modifying agent, resulting in enhanced oil recovery efficiency. The development of water-soluble polymers leads to an increase in the viscosity of water, which makes it easier for carbon dioxide to be absorbed. This was accomplished by reducing the mobility of the gas and delaying its premature release from the solution. By introducing nanoparticles that interact with the polymer chain and create a steric barrier, thereby increasing the CO_2 absorption by ~20%, the system can be enhanced. The polymer enhanced the solution stability of the NPs. As the confining pressure increased and NPs were added, the capacity of the solution to absorb CO_2 also increased. In brief, the incorporation of carbon dioxide into the polymeric nanofluid led to the acquisition of features that enhanced the collection of oil from the reservoir. The interfacial tension can be reduced from ~47 to ~40 mN/m for PAM and from ~36 to ~31 mN/m by simply introducing SiO_2 nanoparticles into the solution. After carbonization, it was further reduced to ~18 mN/m. Using an optimized concentrated carbonated nanofluid for flooding resulted in greater oil output due to the superior oil mobilization of the suspensions compared to PAM and hydrolyzed PAM with nanoparticle combination.

This study presents novel insights into the role of SiO_2 nanoparticles in enhancing the performance of polymeric nanofluids for CO_2 absorption, oil swelling, and oil recovery. The carbonated slug is a cutting-edge technology for the oil and gas industries and is the best option for enhanced oil recovery.

Keywords: Enhanced Oil Recovery (EOR), Polymeric Nanofluid, Carbonated Water Flooding, Polyacrylamide (PAM), SiO_2 nanoparticles

Introduction

Enhanced Oil Recovery, often referred to as EOR, is a significant technical innovation in the field of petroleum engineering. Its purpose is to recover the residual oil that remains after the completion of the primary and secondary recovery procedures. Enhanced oil recovery (EOR) methods have become increasingly important in the petroleum sector as traditional oil reserves decline, and global energy demand continues to rise (Chaturvedi & Sharma, 2020; Halari, Atal, et al., 2025). Compared to the average recovery rates of 30-40% that are obtained by traditional techniques, these enhanced recovery technologies have the potential to significantly improve the efficiency of hydrocarbon extraction, resulting in recovery rates that may reach as high as 60% (Muggeridge et al., 2014).

Carbon dioxide (CO_2) flooding has emerged as a significant EOR method, in which CO_2 injection reduces oil viscosity through miscibility effects while maintaining reservoir pressure. In addition, CO_2 has drawn special attention owing to its unique properties, like being easily soluble in hydrocarbons, potential benefits, and its contribution to global warming (Almobarak et al., 2021; Bisweswar et al., 2020). Field challenges during conventional CO_2 flooding and limited availability make carbonated water flooding a promising technique for enhancing oil recovery from declining oil fields.

Carbonated water injection, also known as CWI, is a recently established cutting-edge recovery method. This technology relies on the interaction between fluids and rocks to function. The use of CWI has the potential to reduce CO_2 leakage during storage, and that caused by buoyancy (Bisweswar et al., 2020; Halari, Bardhan, et al., 2025; Halari et al., 2024). According to the findings of several studies, the implementation of CWI results in a relatively greater amount of carbon dioxide being absorbed by the core minerals of the reservoir compared to alternative methods for tertiary recovery (Ayumu & Babadagli, 2025; Kelemen et al., 2019).

The primary reason for oil recovery is the mass transfer of carbon dioxide (CO_2) that occurs between crude oil and carbonated water, which ultimately leads to the mobilization of oil. Due to the oil swelling, viscosity decrease, and reduction in interfacial tension caused by the dissolved CO_2 in the crude oil, the oil's mobility is increased (Drexler et al., 2020; Seyyedi et al., 2017; Sohrabi et al., 2011, 2012). CWI prevents viscous fingering, minimizes the amount of gravity separation that occurs, and enhances the efficiency with which sweeps are performed compared to traditional water and CO_2 flooding (Chen et al., 2021; Halari, Bardhan, et al., 2025; Honarvar et al., 2017).

The limitations of the lone CWI techniques can be overcome by utilizing various chemical agents, including surfactants, polymers, and nanoparticles. Polymers such as polyacrylamide (PAM) and hydrolyzed polyacrylamide (HPAM) are traditionally used to increase water viscosity and improve sweep efficiency in the reservoir. The long-term performance of nanotechnology has attracted researchers to utilize various EOR techniques (Chaturvedi & Sharma, 2020; Kesarwani et al., 2021; Pandey et al., 2023, 2024; Yadav et al., 2025).

Additionally, it has been demonstrated that the employment of nanoparticles (NP) contributes to the improvement of properties, including an increase in capillary number, a reduction in viscous force and IFT, and a modification of the rock's wetting characteristics. Owing to their smaller size, nanoparticles can penetrate deeper into the reservoir (Sun et al., 2023). Additionally, having a large surface area serves to reduce interfacial energy and provide an effective disjoining pressure, thereby modifying the sandstone's wettability (Halari, Atal, et al., 2025; Sun et al., 2023). Despite their distinct features, the petroleum industry is particularly concerned with stabilising them in the injecting slug. Polymer additives may be the most suitable for stabilizing NPs in the slug, which could potentially aid in recovering more oil from the reservoir. Not only does it facilitate the production of homogeneous polymeric nanofluids (PNF), but it also aids in modifying the viscosity of the polymer slug. As a result of this activity, a combination slug has a greater capacity for the enhancement of all properties that are essential for the EOR.

The incorporation of carbon dioxide (CO_2) into water for enhanced oil recovery (EOR) is a novel technique. The most significant research need for this technology is to increase the CO_2 loading capability of the slug. Chemical agents have been used to enhance CO_2 loading (Chen et al., 2021; Sun et al., 2023). These agents have the potential to enhance CO_2 loading, and in addition, they could retain CO_2 inside the slug, resulting in a greater quantity of CO_2 being trapped in the solution. When it comes to absorbing carbon dioxide in the slug, the most effective additions are polymer and nanoparticles. Additionally, since these chemical additions reduce the interfacial energy, the carbon dioxide that has been entrapped in the slug can be transferred directly into the hydrocarbon (Chaudhry et al., 2024). As a result, the crude oil's viscosity and density decreased. The presence of CO_2 causes the size of the trapped oil to increase, and over time, this trapped oil becomes disengaged from the sandstone surfaces or closed capillaries. Furthermore, nanoparticles (NPs) can generate disjoining pressure, reduce the interfacial energy, and modify the wetting behavior of sandstone. All crude oil, which has low viscosity and density and has been separated from the reservoir, is removed by the polymer present in the slug. This helps improve the mobility control in the reservoir. It was demonstrated that the combined triple-component system was the most effective combination for maximizing the oil recovery factor using this method.

The purpose of this study was to evaluate the synergistic impact between SiO_2 nanoparticles and carbon dioxide in polymer solutions based on PAM and HPAM. Furthermore, the critical properties of PNFs for enhanced oil recovery (EOR) objectives are examined. These factors include the high-pressure IFT, wettability, CO_2 absorption, and viscosity. An innovative carbonated polymeric nanofluid solution has been introduced as a cutting-edge technology for enhanced oil recovery. This was the overriding goal of the present study.

Experimental Section

Materials

Both the water-soluble polymer Pusher 1000 (PAM-polyacrylamide) and hydrolyzed polyacrylamide (HPAM) were acquired from SNF FLOPAM India Pvt., Ltd. The silicon dioxide (SiO_2) nanoparticles were sourced from Sisco Research Lab in India. All tests were conducted using standard deionized (DI) water. For the incorporation of nanoparticles into water, a probe sonicator was used to sonicate the water, resulting in a homogeneous solution. A magnetic stirrer equipped with a Remi Lab 10MLH PLUS was used to dissolve the polymer in water. Absorption studies were conducted using CO_2 with a purity level of 99.99%, provided by Sigma Gases and Services, located in Delhi, India. HPCL – Mittal Energy Limited (HMEL) supplied the crude oil received. The essential physical qualities of crude oil are included in Table 1.

Table 1—Core & Crude Oil Properties

Core DATA		Crude Oil DATA	
Name	Buff Barea	Density (g/cc)	0.94
Diameter (cm)	3.8	API Gravity (°)	18.58
Length (cm)	5.5	Pour Point (°C)	8
Bulk Volume (cc or ml)	62.40	Saturates (% w/w)	54.10
Pore Volume (cc or ml)	7.46	Aromatics (% w/w)	29.28
Porosity (%)	11.96	Resins (% w/w)	7.40
Gas Permeability (mD)	~1200	Asphaltenes (% w/w)	9.22

Methods

First, the commercial SiO_2 NPs concentration was optimized. For this, NPs were taken at various concentrations (250-1500 ppm) and introduced into DI water. The resulting solution was sonicated for one hour to obtain a homogeneous solution. The prepared nanofluid solutions were used to determine the interfacial tension with crude oil. After optimization of the SiO_2 NPs, polymeric nanofluid (PNF) solutions were prepared using PAM and HPAM polymers. To prepare the PNFs, after sonication and the solution had reached homogeneity as a nanofluid, 1000 ppm of polymer was introduced and stirred at 600 rpm for 6-8 hours. Complete hydration of the polymer within the nanofluid. The prepared PNFs were used for further studies, such as contact angle measurements, CO_2 absorption, high-pressure IFT, and oil recovery analysis. The detailed workflow of this study is shown in Fig. 1.

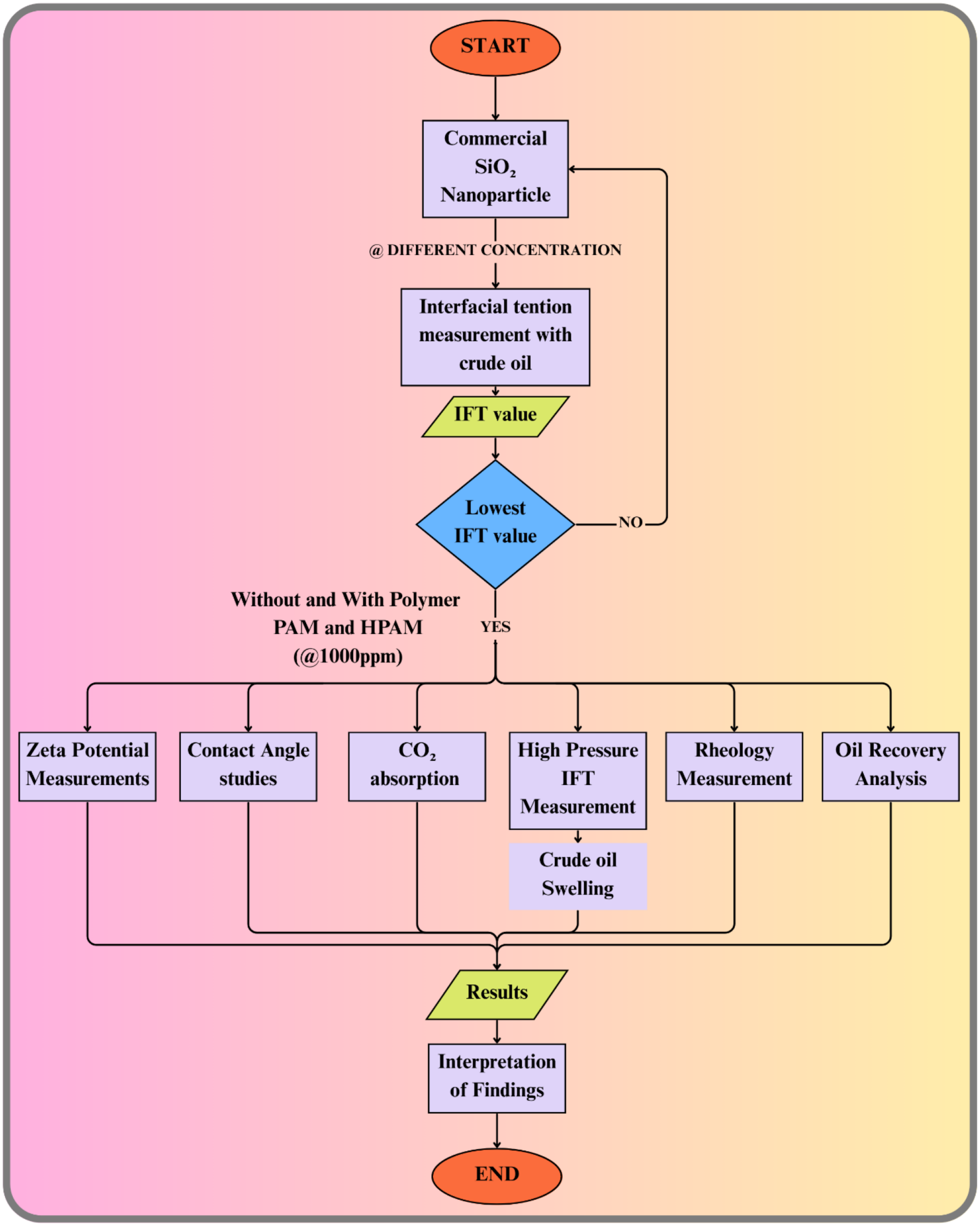

Figure 1—Workflow Diagram

Optimization via IFT Measurement. The optimization of SiO_2 NPs was determined through IFT measurements using the Du Nouy ring technique. The DY-500 tensiometer from KYOWA Scientific was used. This technique involves applying a controlled force to an object with a specific shape between the two liquids and measuring the force required to detach it from the lower liquid phase, thereby determining the interfacial tension. The amount of force, denoted as F, which is needed to lift the ring off the surface of the liquid, was determined. From this measurement, the interfacial tension σ between two immiscible fluids was determined using Equation 1.

$$F = W_{ring} + 4\pi(r_i + r_o)\sigma \tag{1}$$

where W_{ring} is the weight of the ring, and r_i and r_o are the inner and outer diameters of the ring, respectively. The experiment was repeated three times.

Zeta Potential Measurement. The zeta potentials of the three nanoparticle-polymer combinations of SiO_2 NPs in deionized water, SiO_2 NPs in PAM solution, and SiO_2 NPs in HPAM solution were measured using a Malvern Zetasizer Nano-ZS. Each sample was prepared by dispersing SiO_2 NPs at a fixed concentration in an aqueous medium under constant stirring for 30 min, followed by ultrasonication for 15 min to ensure homogeneous dispersion. PAM and HPAM solutions were prepared at identical polymer concentrations using deionized water, allowing complete dissolution under magnetic stirring. After preparation, the samples were equilibrated at room temperature (25 ± 1°C), and zeta potential measurements were conducted immediately (day 0) and subsequently on days 1, 3, and 5. Each measurement was performed in triplicate to ensure statistical reliability, and the average values along with their standard deviations were reported. The pH of the solutions was monitored and maintained within the range of 8-9 to minimize its influence on the surface charge variation.

Contact Angle Studies. Contact angle experiments are often used to determine the wettability of a flat surface (rock surface), which is a property of fluid sticking to a surface saturated with crude oil. The contact angle was determined using a goniometer acquired from Apex Instruments (India) (Acam-NSC series) (Pandey et al., 2023). This experiment involved creating a PNF solution and using a Hamilton syringe to inject a drop of the solution into the rock surface that had been soaked in crude oil. To assess the effect of SiO_2 nanoparticles, a change in the wetting behavior (contact angle) of the oil-dipped core surface was noted.

CO_2 Absorption Test. A high-pressure cell designed to produce carbonated water was used in the CO_2 absorption experiments. With a total volume capacity of 1300 mL, this cell was designed to be safe and dependable during high-pressure operations by withstanding pressures up to 4000 psi, as shown in Fig. 2. To remove any remaining air or impurities, a measured amount of fluid was added to the cell and then evacuated for ten minutes. After evacuation, the empty cell was maintained at a steady temperature of 30°C while being pressurized with high-purity carbon dioxide gas to a confining pressure of 500 psi. The fluid was agitated at 300 rpm until the gas and liquid phases were balanced. The fluid absorbed most of the CO_2 under the specified circumstances when there was a reduction in pressure within the cell. The ultimate pressure was noted once the equilibrium was reached. Equation 2 then uses the actual gas equation to obtain the amount of CO_2 absorbed (in mols).

$$n_{ab} = \left(\frac{P_1}{Z_1} - \frac{P_2}{Z_2}\right) \cdot V \Big/ RT \tag{2}$$

where n_{ab} denotes the number of moles of CO_2 absorbed, P_1 and P_2 represent the initial and equilibrium pressures (Pa), Z_1 and Z_2 are the compressibility factors of CO_2 at these pressures, V is the volume of the cell, R is the real gas constant (8.314 J/mol·K), and T is the absolute temperature in Kelvin (~303.15 K). This method ensures accurate quantification of gas absorption under controlled thermodynamic conditions.

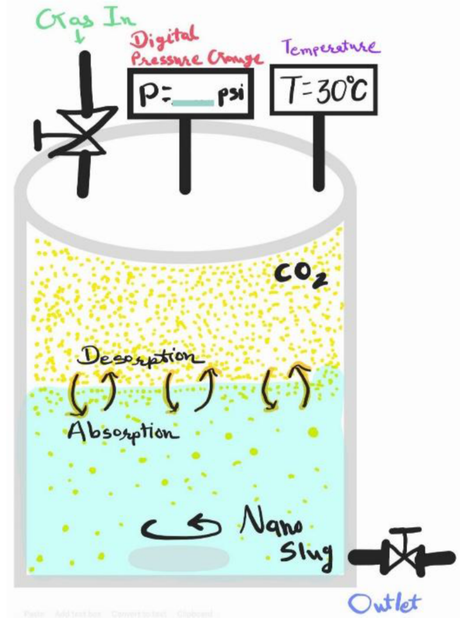

Figure 2—CO2 Absorption setup

High-Pressure IFT Measurement. A high-pressure tensiometer made by DCAM Engineering, Ahmedabad, which uses the pendant/rising drop technique at a confining pressure of 500 psi and a temperature of 30°C, was used to measure the IFT between the solutions and crude oil (Halari, Atal, et al., 2025). The HPHT cell was thoroughly vacuumed before the experiment began, and CO_2 was introduced to maintain a stable pressure. After transferring the prepared carbonated slug to the cell, an oil drop from the bottom was added to compute the IFT at high pressure. Over 15 minutes, the continuous measurement of the IFT was carried out, and the shape and size of the drop were determined. This allowed for the measurement of CO_2 mass transfer and swelling of crude oil upon contact with the carbonated slug.

Rheology Measurement. An Anton Paar MCR 302e rheometer with a double-gap (DG35) configuration was used to perform viscometric evaluation of the manufactured slugs, ensuring good sensitivity across a broad range of shear rates. To assess the shear-dependent rheological behavior of the slugs, the investigation was conducted over a shear rate range of 1 to 1000 s^{-1}. Two experimental settings were used: high pressure (carbonated) and ambient (normal). All the tests were conducted at 30°C. Without any pressure, the slugs were tested in natural settings (normal). Accurate measurement results were obtained for high-pressure experiments by carefully filling the rheometer cell with the slug formulation and then vacuuming to remove any trapped air. A pre-shear at 100 s^{-1} and a confining pressure of 500 psi of the CO_2 gas inside the double-gap geometry were used to create the carbonated slug until pressure equilibrium was reached (~4hrs). Subsequently, regulated pressure conditions were used for rheological testing to evaluate the effect of carbonation on flow behavior. By mimicking the actual operating conditions, this technique enabled a thorough evaluation of rheological reactions in both normal and carbonated settings.

Oil Recovery Analysis. Experiments involving core flooding were conducted to determine whether PNF slugs are beneficial for enhancing oil recovery. The experimental system, built and manufactured by DCAM Engineering in Ahmedabad, consists of a core holder, double accumulators, and a high-precision syringe pump (Halari, Atal, et al., 2025; Halari et al., 2024). All components were designed to ensure that the flow rate remained constant throughout the entire flooding process under the desired pressure conditions. Prior to the studies, the sandstone core samples were thoroughly cleaned using analytical-grade toluene and methanol to eliminate any remaining hydrocarbons and impurities. The cores were then entirely dried by placing them in an oven at 120°C for 24 h. The results of the quantification of petrophysical parameters, including porosity and permeability, are reported in Table 1.

To precisely simulate the reservoir's circumstances, the core holder was subjected to a confining pressure of approximately 3,000 psi, in addition to a back pressure of 250 psi. To determine the initial oil saturation, a period of stabilization lasting twenty-four hours was implemented after the cores had been saturated with crude oil. Water injection marked the beginning of the secondary recovery phase, followed by the tertiary recovery stage, which involved the injection of a carbonated slug with a pore volume (PV) of 0.5. Finally, the phase ended with the injection of the chase water at a flow rate of 0.25 mL/min.

Results and Discussion

This section discusses the potential application of SiO_2 nanoparticles in carbonated slugs to enhance the Oil Recovery from depleted reservoirs.

Nanoparticles optimization through IFT measurement

The nanofluids were prepared at various concentrations of the SiO_2 nanoparticles, and the IFT value with crude oil was determined using the ring method. It is uneconomical to use the nanoparticle at higher concentrations. This can cause an increase in the IFT value, as well as the plugging of microstructural pores within the reservoir (Kesarwani et al., 2021). The optimal concentration of nanoparticles facilitates the optimization of the dosage for maximum IFT reduction during interaction with the crude oil, and the nanoparticles adhere to the interface between the crude oil and nanofluid. SiO_2 nanoparticles exhibit an affinity for the oil-water interface owing to their high surface energy. At low concentrations, nanoparticles tend to migrate to the interface and adsorb, resulting in reduced direct oil-water contact and a decrease in interfacial free energy (Liu et al., 2024; Pandey et al., 2023).

At very low concentrations (up to ~750 ppm) of nanoparticles, owing to insufficient availability of NPs, it is not possible to achieve full interfacial coverage. This may result in a reduction in the IFT but not in the maximum reduction (see Figure 3). For SiO_2 nanoparticles, 750 ppm is the optimized concentration at which maximum interfacial adsorption can occur, resulting in a compact NPs layer. At 750 ppm, the concentration at which the maximum reduction in interfacial tension was observed, the interfacial tension decreased to 14.87 mN/m. Beyond 750 ppm, the interfacial tension increased slightly (see Figure 3) because at higher concentrations of the NPs, interparticle attraction dominates, which may lead to aggregation and sedimentation (Sircar et al., 2022). These bulk aggregates exhibit very limited interfacial activity, which hinders their potential to reduce IFT. From this, a 750 ppm concentration of SiO_2 nanoparticles was optimized, and all subsequent studies were carried out at this concentration.

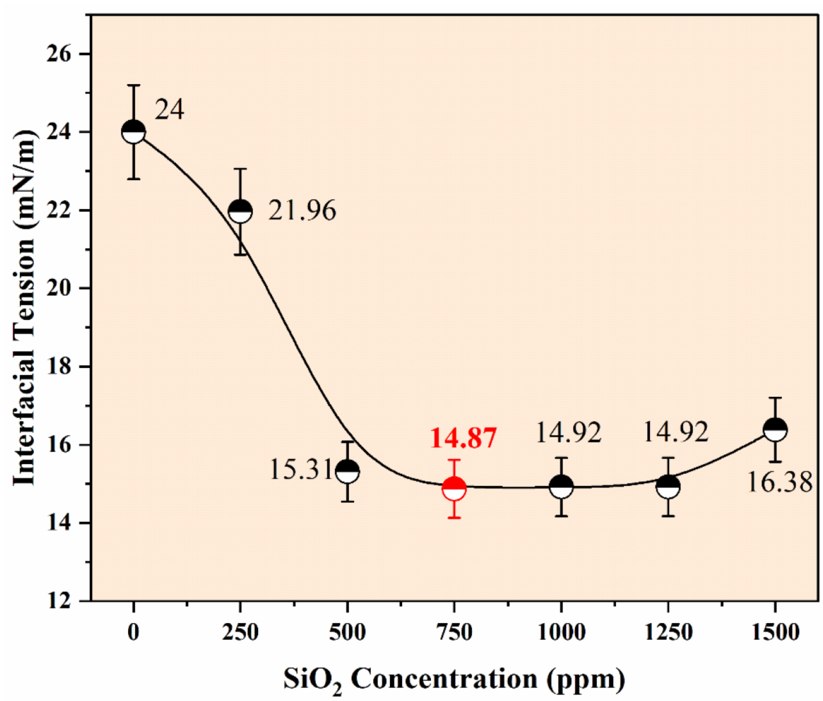

Figure 3—Interfacial Tension measurement at different SiO_2 concentrations

Zeta Potential Measurement

The dispersion stability of nanoparticles in aqueous and polymeric media dictates their efficacy under dynamic reservoir conditions in EOR applications (Zhang et al., 2014). The zeta potential, which serves as a key indicator of the dispersion stability, provides insight into the electrostatic repulsive forces between particles suspended in a liquid. A dispersion is typically regarded as electrophoretically stable when its zeta potential is greater than ±30 mV (Bardhan et al., 2024; Halari, Atal, et al., 2025). The data presented in Figure 4 provide a comparative analysis of the zeta potential of SiO2 NPs in three different aqueous environments: deionized water, PAM solution, and HPAM solution, over a five-day time span.

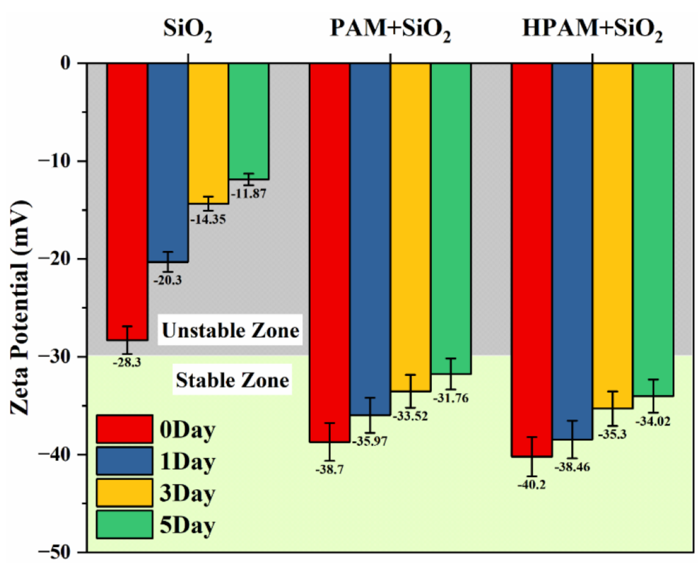

Figure 4—Zeta potential values of PNFs with time

Initially, SiO_2 NPs dispersed in deionized water exhibited a moderately high negative zeta potential of -28.3 mV on day 0, nearing the threshold for colloidal stability. However, this value declined progressively to -20.3 mV, -14.35 mV, and ultimately -11.87 mV by the fifth day, indicating a substantial deterioration in dispersion stability. This temporal trend suggests the rapid agglomeration or sedimentation of SiO_2 NPs in the absence of stabilizing agents, rendering the system unsuitable for long-term EOR applications that require sustained dispersion (Nguyen et al., 2014). In contrast, when SiO2 NPs were introduced into the PAM and HPAM solutions, the zeta potential values were consistently beyond the -30 mV threshold across all time intervals. For PAM+ SiO_2 NPs, the zeta potential initially measured -38.7 mV and demonstrated a mild decline to -35.97 mV, -33.52 mV, and -31.76 mV over five days. Similarly, in the HPAM+ SiO_2 NPs system, the zeta potential started at -40.2 mV and remained relatively stable at -38.46 mV, -35.3 mV, and -34.02 mV, even after five days. These observations confirmed that the inclusion of both PAM and HPAM significantly enhanced the electrophoretic stability of the SiO_2 NPs dispersions. Therefore, the temporal stability exhibited by polymer-assisted dispersions indicates that polymeric additives can mitigate nanoparticle agglomeration. Both PAM and HPAM provide a steric and electrostatic stabilization mechanism, wherein polymer chains adsorb onto the surface of the SiO2 NPs, introducing additional repulsive interactions that hinder inter-particular aggregation (Zhang et al., 2014). Notably, HPAM consistently yielded more negative zeta potential values than PAM, suggesting superior stabilizing capability. This may be attributed to its partially hydrolyzed structure, which imparts a higher density of carboxylate groups, thereby increasing the surface charge density upon adsorption and promoting greater electrostatic repulsion (Adil et al., 2016).

Among the three systems evaluated, the HPAM + SiO_2 NPs demonstrated the most favorable characteristics for long-term deployment in carbonated water-based EOR formulations. It displays the most negative initial zeta potential, indicative of strong inter-particulate repulsion, but it also maintains a relatively consistent profile over five days, signifying minimal loss in dispersion stability. This suggests that HPAM is more effective in preserving the colloidal integrity of the SiO_2 NPs over time, which is crucial for maintaining injectivity and uniform sweep in porous media.

Contact Angle Studies

Contact angle analysis revealed a distinct transition in the surface wettability upon treatment with SiO_2 nanoparticles. The time-resolved contact angle analysis presented here provides a comparative evaluation of the dynamic wettability behavior of various formulations, including the PAM/HPAM polymers and SiO_2 nanoparticle combinations. The contact angle with a 1000 ppm PAM and HPAM solution on an oil-wet surface decreased from 91° to 72° and from 87° to 60°, respectively, and this decrease was accompanied by the introduction of SiO_2 nanoparticles. Figure 5 illustrates that the angle decreased over time owing to the presence of nanoparticles in the solution. Beginning with its original state, the solution that included the nanoparticles had a reduction in its dynamic contact angle after 30 min. The modification of the wettability of the oil-wetting sandstone pellet has been proven to have occurred because of the decrease in the contact angle (Sircar et al., 2022). The wetting behavior of the sandstone rock was altered from intermediate-wet to water-wet conditions because of the addition of nanoparticles to the solution under consideration. After the introduction of SiO_2 NPs, the contact angles of the PNF solutions decreased from 89° to 10° and from 81° to 13° for PAM + SiO_2 and HPAM + SiO_2, respectively.

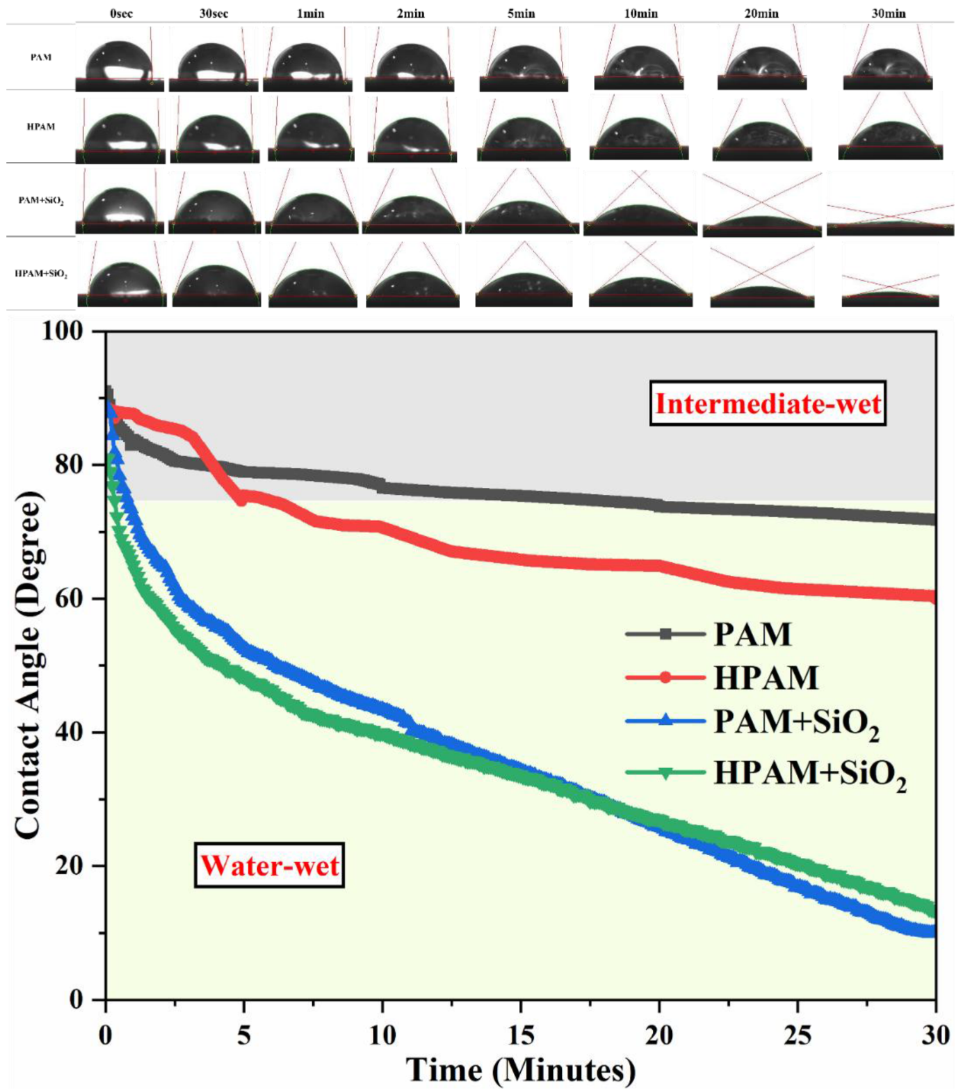

Figure 5—Contact angle images and values with Time

There is a lower surface energy at the interface between the solid and liquid, which makes it more advantageous for an aqueous solution to spread and cover the surface. The nanoparticles can efficiently adjust the surface energy of the sandstone, which in turn reduces the propensity for oil to adhere to the surface and enhances the wetting behavior of the aqueous solution (Dalal Isfehani et al., 2024; Sircar et al., 2022). It is possible for the structural disjoining pressure to eliminate the capillary pressures that were originally responsible for the oil's ability to moisten the sandstone surface. Due to the presence of nanoparticles, the total surface energy is reduced, making it easier for the aqueous solution to interact with and penetrate the surface. Concurrently, the structural disjoining pressure is responsible for aggressively dislodging the oil molecules, which in turn facilitates a shift from an oil-wet state to a more water-wet state (Chaturvedi & Sharma, 2020; Kesarwani et al., 2021; Seyyedi et al., 2017).

CO_2 Absorption

The observations of CO_2 absorption capacity across various formulations provided insightful information about the synergistic behaviour of SiO_2 nanoparticles and PAM/HPAM in facilitating CO_2 solubilization through separate and complementary processes. Carbonic acid is first produced between the solution layer and CO_2 gas when the pressure of the CO_2 gas is supplied to the slug inside the vessel. Increasing the pressure of CO_2 gas in the vessel causes it to dissolve under high-pressure conditions, which results in the production of carbonated fluids. The baseline absorptions in PAM and HPAM were 0.484 mol/kg and 0.502 mol/kg, respectively. This reflects the limited gas solubility in the formulation owing to the inherently low CO_2-solution interfacial area and high surface energy. The primary reason for the higher CO_2 absorption in the HPAM solution was its polymer structure and composition. PAM contains a $CONH_2$ functional group, which makes the solution neutral. However, HPAM is partially converted to the -COO group, which makes the solution slightly basic (Mo et al., 2018). During interaction with CO2, PAM typically interacts and forms a weak acid. However, as the HPAM solution is slightly basic, when it interacts with CO_2, the acid-base interaction occurs, which may result in higher CO_2 absorption (Al-Kindi et al., 2022; Ma et al., 2015; Mo et al., 2018).

The addition of nanoparticles to the slug can reduce the IFT and enhance the CO_2 absorption capacity, as illustrated in Figure 6. The introduction of NPs helps lower the interfacial energy of the solution, which may result in a better CO_2 absorption capacity of the PNFs. SiO_2 nanoparticles have a relatively high surface area, which enables the interaction with the CO_2 molecules via hydrogen bonding (Chaturvedi & Sharma, 2020). In addition, it can effectively trap CO_2 molecules within the bulk solution. After adding NPs in the PAM and HPAM solutions, the CO_2 absorption increased to 0.516 mol/Kg (6.51%) and 0.59 mol/Kg (17.34%), respectively (see Figure 7). The HPHT measurement of interfacial tension validates these findings.

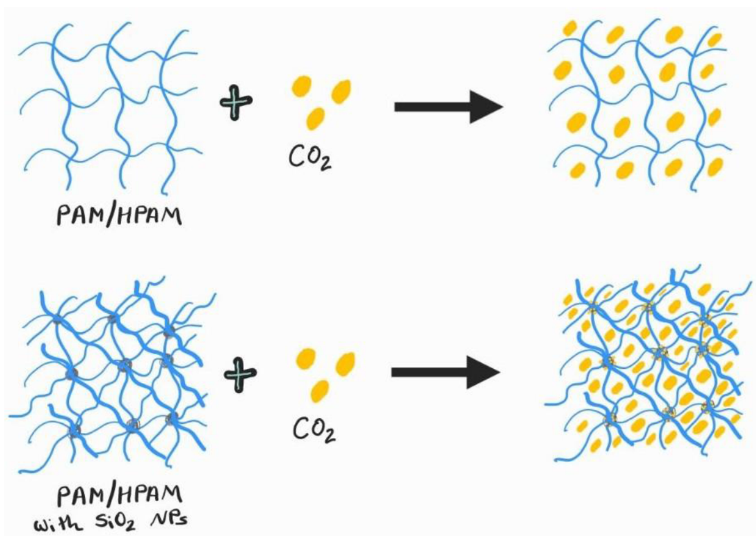

Figure 6—CO2 absorption mechanism in polymer and PNF solution

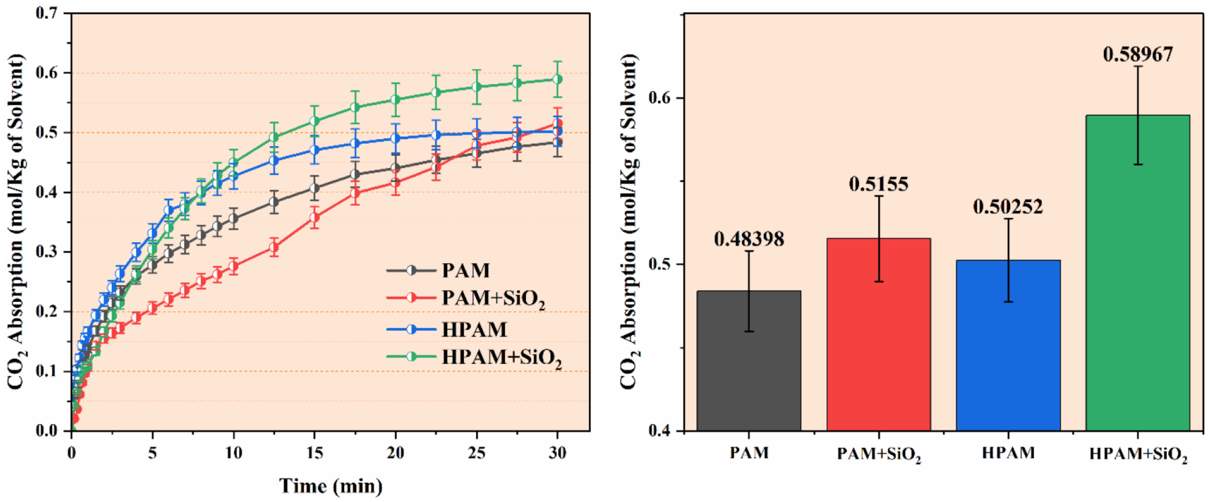

Figure 7—CO2 absorption results with Time

High-Pressure IFT Measurement

The interfacial tension represents the behavior of the prepared solution with the reservoir fluid (crude oil). High-pressure IFT measurements were conducted using the rising-drop method in a closed cell. The IFT reduction between the slug and crude oil is a critical parameter in the EOR process, particularly when targeting capillary-trapped oil. Under normal conditions, the IFT is relatively high, as shown in Figure 8. The incorporation of CO_2 into the system enables the activation of physicochemical interactions at the oil-slug interface, including electrostatic repulsion, steric hindrance, and CO_2 diffusion enhancement, and crude oil swelling (Badmos et al., 2019; Halari, Atal, et al., 2025). The introduction of CO_2 into the PAM and HPAM solutions decreases the IFT values from 47.44 mN/m to 40.51 mN/m and from 35.42 mN/m to 31.09 mN/m, respectively. In addition, NPs can further reduce the systems' IFT to 24.68 mN/m & 21.94 mN/m at normal and 19.23 mN/m & 18.99 mN/m at carbonated conditions for PAM and HPAM PNFs, respectively.

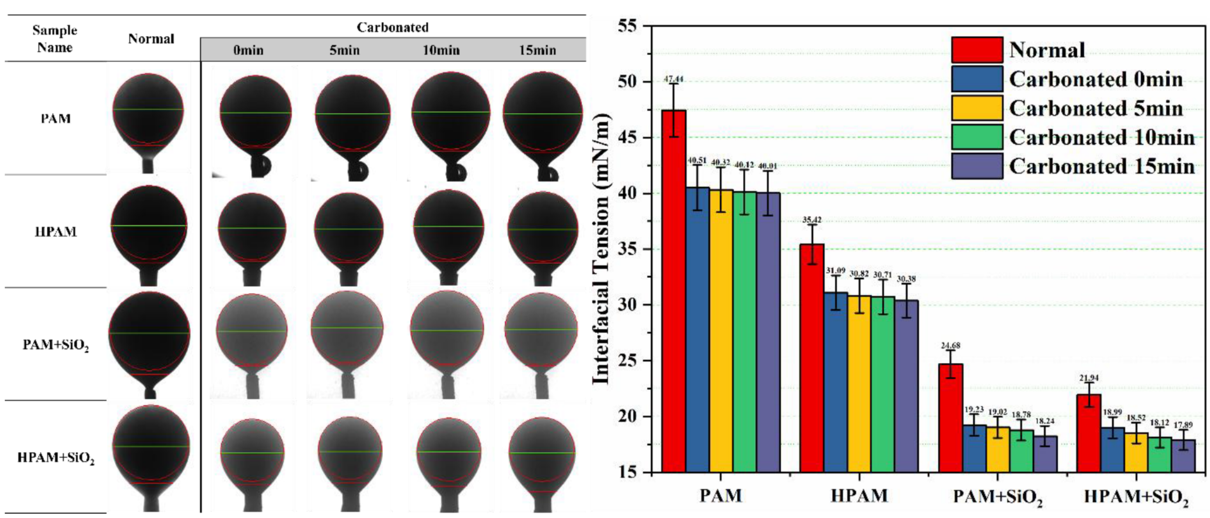

Figure 8—Interfacial tension measurement and drop images at normal and carbonated conditions with Time

Under carbonated conditions, with time, CO_2 is prone to diffuse from the carbonated solution to the crude oil, resulting in minor swelling of the crude oil and further lowering of the IFT. Because CO_2 is highly soluble in hydrocarbons, notable changes in the drop shape and IFT value can be observed (Golkari et al., 2022). Swelling is evident from the image, as indicated by the increasing droplet diameter and reduction in the curvature of the oil drop over time. This CO_2 diffusion/mass transfer resulted in a reduction in the oil viscosity and density. Under reservoir conditions, trapped oil ganglia expand and reconnect with percolation pathways through oil swelling, resulting in snap-off and detachment from the pore walls. Thus, this triple-component system of PAM/HPAM + SiO_2 + CO_2 exhibits the maximum capability for reducing the IFT with crude oil.

Rheology Analysis

Viscosity measurements of the prepared slugs were conducted under normal and carbonated conditions using a rheometer. Continuous measurements of viscosity after applying CO_2 to the solution provide a real-time analysis of the solution and its behavior as it becomes carbonated. Carbonation is a dynamic process in which absorption and desorption rates change over time. The solution with a high CO2 absorption capability exhibited the highest absorption, and a sudden drop in viscosity was observed, as shown in Figure 9. For the PAM + SiO_2 and HPAM + SiO_2 PNFs, a sudden drop in the viscosity indicates a higher absorption rate than that of the PAM and HPAM solutions, respectively.

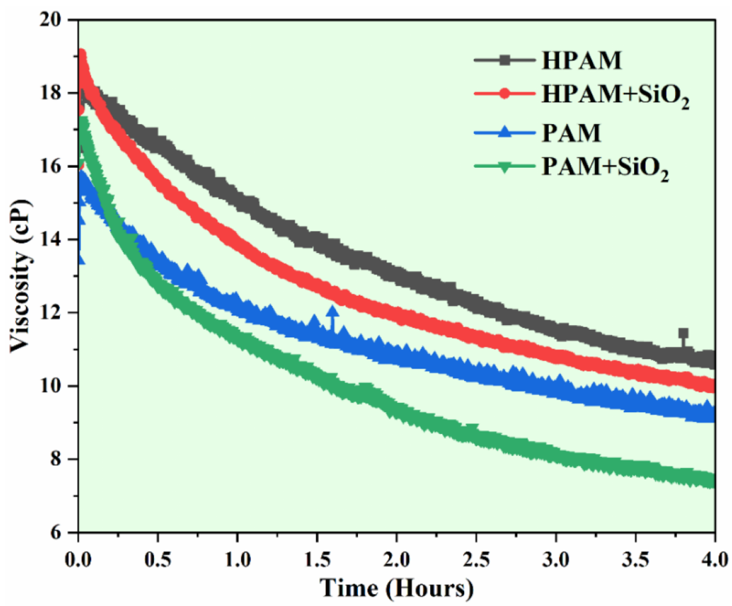

Figure 9—Viscosity profile vs Time during carbonation process

The viscosity of the polymer slug at a concentration of 1000 ppm decreased, resulting in an increase in the shear rate from 1 to 1000 s^{-1}. The trend in the graph illustrates the pseudoplastic behavior of the solution via its behavior. The observed drop in viscosity may be attributed to the weak entanglement of a vast and intricate polymeric chain, which can break at a higher shear rate (Halari, Atal, et al., 2025; Kesarwani et al., 2021). This might be the cause of the observed decrease in the resistance. Nanoparticles have a more significant impact on the viscosity of a solution when added to it. Researchers have observed that the viscosity of polyacrylamide polymer solutions increases when hydrophilic nanoparticles are added to the solution (Sircar et al., 2022). The internal molecular structure of the material is responsible for this behavior, characterized by the fact that when the material is at rest, long filamentary molecules of uncrosslinked polymers compress to form balls. These balls undergo a transformation into an ellipsoidal shape when subjected to shear conditions, resulting in an increase in the disentanglement of the molecules (Halari, Atal, et al., 2025; Sircar et al., 2022). When these polymers are subjected to carbonation, the CO_2 molecules reside between the polymer chain gaps, as shown in Figure 6. The addition of SiO_2 nanoparticles increases the polymer viscosity by forming a bridge between the two polymer chains, thereby partially cross-linking them (Chaturvedi & Sharma, 2020). CO_2 intrusion into the solution slightly decreased the viscosity profile of the solution, as it formed in situ carbonic acid and reduced the density of the fluid. Disruption is caused by the presence of carbon dioxide in the solution, which affects the interaction between the SiO_2 nanoparticles and polymer chains. Owing to this event, the chain bridges of the polymer undergo partial disintegration, resulting in a decrease in the overall viscosity of the material (see Figure 10). At an initial shear rate of 1 s^{-1}, the viscosities of PAM, HPAM, PAM + SiO_2, and HPAM + SiO_2 were measured to be 440, 320, 470, and 489 cP, respectively. After carbonation, the initial viscosity values of the samples decreased to 180, 215, 126, and 105 cP.

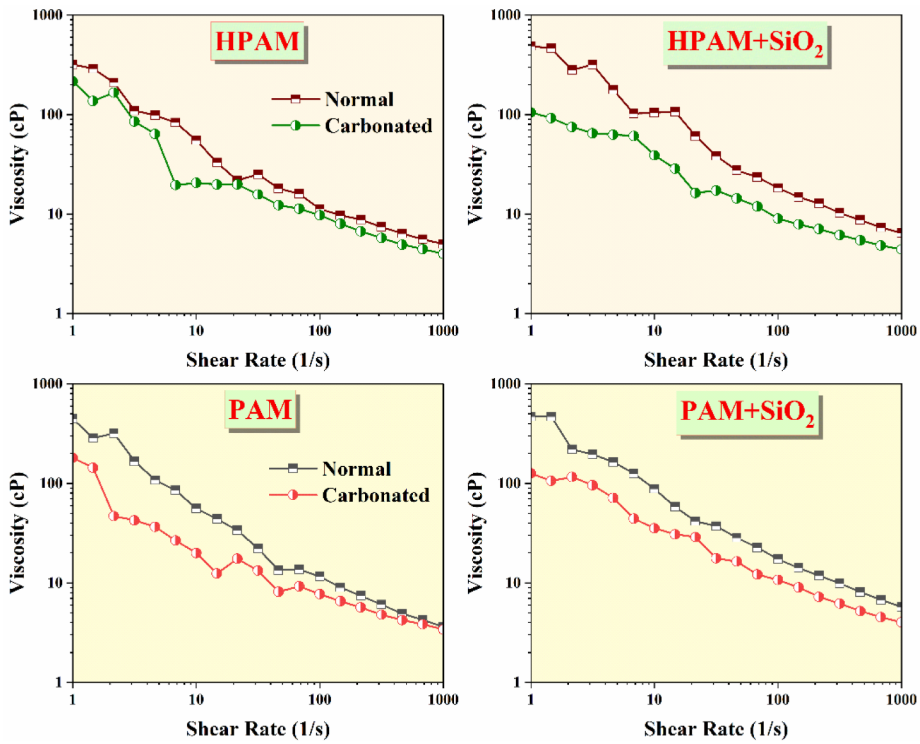

Figure 10—Viscosity profile over varying shear rate at normal and carbonated conditions

Flooding results

Core flooding tests were conducted under carbonated conditions to evaluate the oil recovery efficiency of the slugs. These studies were conducted both in isolation and in conjunction with a polymer (PAM/HPAM) and SiO_2 nanoparticles. In the first step of the process, the sandstone core was flooded with water to investigate liquid permeability. Crude oil was poured into the core. After the first oil saturation was achieved, the soil was kept undisturbed for a period of twenty-four hours. The purpose of this aging process is to enable the core sample to achieve stable conditions, at which point the interaction between the injection oil and pore structure of the rock becomes indicative of the actual reservoir conditions. Initially, water flooding was used as a secondary recovery strategy until the oil production ceased. Immediately after that, 0.5PV of the carbonated PNF slug was injected into the sandstone core, and then chase water flooding was carried out until the oil stopped flowing from the core. Table 2 lists the different oil recovery parameters encountered during flooding.

Table 2—Oil Recovery parameters

Chemical Composition of Slug	Saturation (%)			Secondary Oil Recovery (% OOIP)	Additional Oil Recovery (% OOIP)	ΔP (psi)	Cumulative Oil Recovery (% OOIP)
	S_{oi}	S_{wi}	S_{or}				
C-PAM	92.59	7.41	55.07	40.52	8.83	5	49.35
C-HPAM	91.11	8.89	53.6	41.18	11.76	7	52.94
C-PAM + SiO_2	92.72	7.28	54.26	41.47	20.66	12	62.14
C-HPAM + SiO_2	91.78	8.22	53.6	41.61	27.74	13	69.34

It has been noticed that the oil recovery factor is greatest for the carbonated HPAM + SiO_2 slug. This was confirmed by verifying the findings of high-pressure IFT and contact angle measurements obtained previously. The percentage of the recovery factor is shown in Figure 11 as a function of the injected pore volume of the fluids, including carbonated PNF solutions. The pressure drop encountered during the flooding process indicates the ease or difficulty of slug flow through the sandstone core. Comparing all flooding results, a maximum pressure drop of 13 psi occurred during HPAM + SiO_2 solution flooding, which gave the best recovery factor of 69.34%. Viscometry analysis also corroborated these results. A larger pressure drop in the PNFs resulted in an increase in the areal coverage of the solution. It is anticipated that this will lead to a stronger interaction with crude oil, ultimately resulting in a higher mobilization of oil from the core.

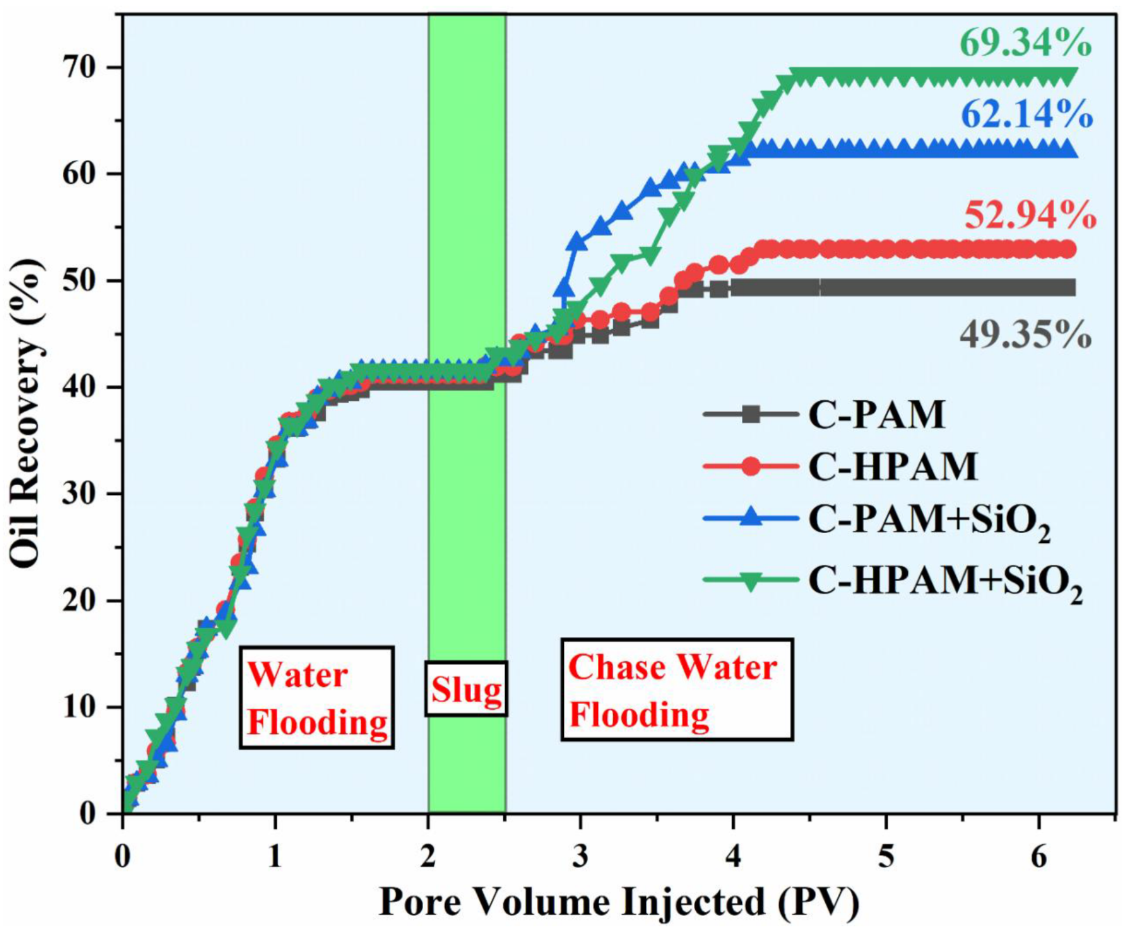

Figure 11—Oil recovery factor vs injected pore volume graph

Energy Economics

The limited availability of CO_2 presents a strong case for the commercialization of Carbonated Water Injection (CWI)-based EOR technology. Compared to conventional CO_2 flooding and CO_2-WAG, CWI offers distinct advantages in cost-effectiveness, operational simplicity, and infrastructure compatibility. Since it requires only minor modifications to existing flooding systems, CAPEX is reduced by nearly 50%. By lowering CO_2 demand, CWI also cuts operating costs and shortens the payback period by 30-40%, making projects more economically viable. Unlike direct CO_2 injection, which accelerates equipment corrosion and necessitates costly facility upgrades, CWI mitigates such risks through dissolved CO_2 injection, thereby extending the life of surface facilities. Thus, CWI emerges as a technically feasible and commercially sustainable alternative, with strong potential for large-scale deployment in oil recovery under CO_2-limited scenarios.

Conclusion

Regarding the use of silicon dioxide nanoparticles in the EOR process involving carbonated fluids, a comprehensive study was conducted to address the current research gap and expand the knowledge base. This study provides vital insights that have the potential to revolutionize the industrial approach to increasing oil recovery. These insights will be gained through the analysis of zeta potential, wettability modifications, and interfacial activity assessment, as well as rheological features and performance of polymeric nanofluid flooding. Obtaining a more optimized and efficient EOR process is feasible using SiO_2 nanoparticles loaded with carbon dioxide. This development will ultimately lead to an increase in oil output as well as significant economic benefits. The results of this study may have made this possible. When an optimal concentration of SiO_2 (750 ppm) was added to the PAM/HPAM polymer solution, a significant shift in the stability of the nanoparticles was observed. In addition, the ability to load CO_2 is increased by 17.34% in HPAM, while it is increased by 6.51% in PAM.

As an additional point of interest, the analysis of the changes in wettability that occur as a consequence of introducing SiO_2 nanoparticles will provide insights into the potential of these nanoparticles as agents for enhancing oil recovery. The use of HPAM resulted in an improvement in the interfacial properties in contrast to the use of PAM. Although the IFT decreased to ~18 mN/m (C-HPAM + SiO_2), other values, such as ~40 mN/m (C-PAM), ~31 mN/m (C-HPAM), and ~19 mN/m (C-PAM + SiO_2), were found to be higher. This phenomenon was observed after carbonation of the solutions or addition of CO_2. Because the viscosity profile of HPAM + SiO_2 was higher than that of the other PNFs, a larger differential pressure was generated across the core during the oil recovery process. This was because 27.74% (C-HPAM + SiO_2) tertiary oil recovery was achieved. This is a larger percentage than those of 8.83% (C-PAM), 11.76% (C-HPAM), and 20.66% (C-PAM + SiO_2) that were achieved. The superior results achieved were due to the unique qualities of the SiO_2 nanoparticles and hydrolyzed PAM polymers. These characteristics are beneficial for improving all attributes. When examined as a unit, the study of carbonated polymeric nanofluid flooding will enable an understanding of the efficacy and efficiency of this technology in the specific context of the EOR.

Declaration of interests

The authors declare that they have no known competing financial interests or personal relationships that could have influenced the work reported in this study.

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